Highlights

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Characterization Study

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- Decoder contribution in 3D medical image segmentation is spatially heterogeneous.
- Matched full and efficient decoders exhibit net-neutral global voxel correctness.
- Over 80% of positive decoder benefit is concentrated within 5% of the volume.
- Inference-time uncertainty predicts flip location but not improvement direction.

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Characterization Study

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Abstract

U-Net based 3D medical image segmentation models increasingly incorporate heavy decoders, including multi-scale attention and Transformer modules. Efficient alternatives simplify or remove these decoder stages based on global end-point metrics, assuming that static decoder capacity is uniformly redundant. However, where and under what spatial conditions decoder computation actually contributes remains uncharacterized. In this work, we present a systematic characterization study of decoder computation in 3D medical image segmentation. By evaluating voxel-level prediction changes between matched full and efficient decoders across representative architectures and datasets, we analyze decoder utility in terms of spatial heterogeneity, benefit concentration, and inference-time predictability. Our findings show that decoder contribution is spatially heterogeneous: full decoders yield a statistically net-neutral global benefit, as positive corrections are offset by negative degradations. This net-neutral effect is tightly localized within a thin boundary band, demonstrating structured redundancy rather than global irrelevance. Furthermore, while lightweight inference-time uncertainty signals reliably identify locations of prediction instability, they fail to predict the direction of improvement, rendering simple selective-allocation strategies ineffective. This study shifts the evaluation of decoder efficiency from architectural modification toward fine-grained computation characterization,

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establishing empirical constraints for future adaptive inference designs.

Keywords: 3D Medical Image Segmentation, Efficient Deep Learning, Uncertainty Quantification, Decoder Redundancy, Model Characterization

1. Introduction

Precise 3D medical image segmentation is essential for clinical tasks such as surgical planning, tumor monitoring, and radiation therapy. In deep learning architectures, encoder-decoder networks based on the U-Net paradigm [1, 2] remain the foundational standard. Recent research has primarily focused on developing highly expressive feature encoders, using vision transformers and large convolutional kernels [3, 4, 5, 6]. In parallel, modern 3D architectures retain heavy, multi-resolution decoders equipped with complex skip-fusion and refinement modules. This architectural trajectory assumes that dense decoder capacity is universally necessary across the image volume.

However, standard evaluation paradigms assess decoder designs almost exclusively through aggregate metrics such as the global Dice similarity coefficient or Hausdorff distance. While these macroscopic metrics summarize overall performance, they obscure the spatial distribution of prediction changes. Consequently, a basic scientific question remains open:

How does additional decoder computation contribute to 3D medical image segmentation?

To address this question, we present a characterization study of decoder computation in 3D segmentation models. Rather than introducing a new network architecture, we advocate computation characterization as a research approach complementary to architecture design. We utilize matched pairs of full and efficient decoders (building on the EffiDec3D framework [7]) as controlled experimental probes. By holding encoder weights, training protocols, and data splits fixed, we isolate decoder capacity and directly observe voxel-level prediction shifts (referred to as "flips") induced by decoder scaling (Figure 1).

We focus specifically on the decoder pathway as a well-defined subsystem responsible for spatial resolution reconstruction and multi-scale feature fusion. This focus allows controlled causal attribution while leaving feature-extractor encoders fixed. Skip-feature fusion is treated as part of the decoder pathway, whereas encoder representation dynamics present separate research questions requiring distinct experimental controls. Unlike efficient architecture papers that optimize for speed, our primary objective is to understand computation itself.

We characterize decoder computation across three complementary dimensions: (i) spatial heterogeneity, (ii) benefit concentration, and (iii) inference-time predictability.

The remainder of this paper is organized as follows: Section 2 reviews related literature and highlights key gaps. Section 3 describes our evaluation framework and metrics. Section 4 details our experimental results. Section 5 discusses underlying mechanisms and practical implications, and Section 6 concludes the paper.

2. Related Work

2.1. Decoder Design in 3D Medical Image Segmentation

The encoder-decoder architecture introduced by U-Net [1, 2] and V-Net [8] remains the central design pattern for 3D medical image segmentation. Frameworks such as nnU-Net [9] demonstrated the effectiveness of standardized training protocols combined with deep decoding pathways. Subsequent developments incorporated Transformer backbones and large-kernel convolutions to capture long-range contextual dependencies (e.g., TransUNet [10], UNETR [3], Swin UNETR [4], 3D UX-Net [5], and MedNeXt [6]). These modern architectures emphasize expressive feature extraction while maintaining dense multi-scale decoders. However, these designs are evaluated primarily through final segmentation accuracy and overall parameter count; how decoder capacity operates spatially across volume regions remains unexamined.

2.2. Efficient Segmentation Architecture Design

To mitigate the high computational cost of 3D volumetric convolutions, recent works have explored lightweight decoder designs. EffiDec3D [7] introduced systematic decoder pruning by reducing channel dimensions and eliminating high-resolution decoding stages, establishing trade-offs between parameter scale and Dice performance. Similarly, EfficientMedNeXt [11] simplified decoding stages before introducing residual blocks. Other lightweight models achieve efficiency through hybrid transformer-convolution encoders, attention pruning, or custom lightweight modules (e.g., SegFormer [12], UNETR++ [13], Slim UNETR [14]). Nevertheless, efficient architecture studies infer redundancy indirectly from global performance curves, leaving unexamined whether the eliminated computation was spatially uniform or concentrated in specific anatomical sub-regions.

2.3. Model Characterization and Uncertainty Quantification

Model characterization studies in medical imaging have focused on uncertainty estimation, calibration, failure detection, and quality assessment using techniques such as Monte Carlo Dropout [15] and Deep Ensembles [16] (e.g., Jungo & Reyes [17], recent CMIG studies [18, 19], and comprehensive reviews [20]). These works analyze how uncertainty maps correlate with segmentation errors or clinician confidence. However, existing characterization frameworks assess general prediction uncertainty rather than the marginal spatial utility of decoder computation. The relationship between spatial uncertainty and decoder capacity remains unexplored.

3. Methods: A Characterization Framework

To understand the spatial and functional utility of decoder computation, we design a controlled observation study. We treat a matched full decoder and an efficient decoder as experimental instruments, isolating the effect of decoder capacity on voxel-level predictions.

We define decoder computation as the sequence of upsampling, skip-feature fusion, and post-fusion refinement operations that transform the bottleneck representation into dense high-resolution predictions. In our experimental design, the encoder feature extractor is held fixed, architecture-specific skip features are held fixed, and interventions occur strictly on decoder width (channel count) and depth (stage removal). For stage removal interventions, the associated skip-fusion operations are treated as part of the removed decoder pathway.

3.1. Experimental Setup

Let v denote a voxel within a 3D medical image. We evaluate two decoder configurations sharing identical encoder features, training protocols, and data splits:

- **Efficient Decoder (E_1)**: Parameter-matched efficient decoder based on the EffiDec3D configuration, utilizing concatenation for skip aggregation to strictly align with published parameter counts.

- **Full Decoder (E_0)**: The standard dense, high-resolution decoder of the respective backbone architecture.

For a given voxel v, let $y(v)$ be the ground truth label, $\hat{y}_e(v)$ be the prediction from the efficient decoder, and $\hat{y}_f(v)$ be the prediction from the full decoder.

3.2. Voxel-level Flip Analysis

Rather than relying on aggregate metrics like Dice, which sum all errors indiscriminately, we track how the full decoder alters the prediction of the efficient baseline. We define three categories of prediction "flips" when transitioning from the efficient to the full decoder:

- **Positive Flip ($P(v)$)**: The full decoder improves a previously incorrect prediction. $P(v) = 1[\hat{y}_e(v) \neq y(v) \wedge \hat{y}_f(v) = y(v)]$.
- **Negative Flip ($N(v)$)**: The full decoder degrades a previously correct prediction. $N(v) = 1[\hat{y}_e(v) = y(v) \wedge \hat{y}_f(v) \neq y(v)]$.
- **Net Benefit ($U(v)$)**: The voxel-wise difference between improvements and degradations. $U(v) = P(v) - N(v)$.

The global rates are defined as $R_{pos} = \text{mean}_v P(v)$, $R_{neg} = \text{mean}_v N(v)$, and $R_{net} = R_{pos} - R_{neg}$.

3.3. Core Metrics

We operationalize our investigation into structured redundancy through the following specific metrics:

1. Global Net Benefit: This metric evaluates the macro-level necessity of the full decoder. The global net flip rate R_{net} is computed as the expected voxel-wise net benefit over the volume V:

$$R_{net} = \frac{1}{|V|} \sum_{v \in V} U(v) \tag{1}$$

Its physical meaning answers whether the full decoder provides a unidirectional advantage ($R_{net} > 0$) or if its positive corrections and negative degradations cancel each other out ($R_{net} \approx 0$).

2. Boundary and Anatomy Resolution: To evaluate spatial heterogeneity, we analyze the net benefit $U(v)$ as a function of the Euclidean distance to the nearest organ boundary, as well as its distribution across different anatomical structures. This reveals *where* the decoder is actually intervening.

3. Oracle Concentration: This metric measures the spatial sparsity of the decoder's utility. We divide the image into 3D blocks b_r and compute the net benefit $B(r) = \sum_{v \in b_r} U(v)$ for each block. We rank the blocks by $B(r)$ and calculate the fraction of the total positive net mass recovered by the top $k\%$ of blocks:

$$Coverage(k) = \frac{\sum_{r=1}^{k\% \times |B|} \max(B(r), 0)}{\sum_{r=1}^{|B|} \max(B(r), 0)} \tag{2}$$

We summarize this curve using the K_{80} metric, which is the execution budget required to recover 80% of the full decoder's positive benefit.

4. Flip Predictability: This evaluates whether we can forecast the decoder's behavior before executing it. We use lightweight inference-time signals from the efficient decoder, such as predictive entropy:

$$H(v) = -\sum_{c=1}^{C} p_c(v) \log p_c(v) \tag{3}$$

where $p_c(v)$ is the softmax probability for class c. We measure predictability using the Area Under the Receiver Operating Characteristic (AUROC) to predict the *location* of flips (instability) and the *direction* of flips (positive vs. negative).

5. Selection Gap: This measures the practical, deployable opportunity for selective computation. We compare the net benefit recovered by an uncertainty-based block selector (e.g., selecting blocks with the highest mean entropy) against a random selection policy at a fixed computational budget (e.g., 20%). A significant positive gap indicates a viable routing strategy.

4. Experiments and Results

4.1. Experimental Setup and Baseline Reproduction

We evaluate matched decoder pairs across three representative 3D medical image segmentation architectures: a large-kernel CNN (3D UX-Net [5]), a vision transformer backbone (SwinUNETR [4]), and a ConvNeXt-style network (MedNeXt-M-K3 [6]). Experiments are performed on the BTCV multi-organ

Table 1: Quantitative comparison of Full (E_0) vs. Efficient (E_1) decoders on the BTCV 13-organ dataset across three backbones.

Architecture	Variant	Params (M) ↓	FLOPs (GMac) ↓	Latency (ms) ↓	Mean Dice (%) ↑	Mean HD95 (mm) ↓
3D UX-Net	Full (E_0)	53.01	578.74	40.6	79.18	9.04
	Effi (E_1)	3.16	49.49	8.0	77.73	11.82
SwinUNETR	Full (E_0)	62.19	308.24	37.0	80.48	8.88
	Effi (E_1)	11.21	55.84	16.7	79.49	8.90
MedNeXt-M-K3	Full (E_0)	39.32	230.87	40.9	83.74	5.77
	Effi (E_1)	12.95	106.91	18.7	81.47	6.60

CT dataset (13 organs) [21] and the FeTA 2021 fetal brain MRI dataset (7 structures) [22].

Hardware and Training Protocol: Models were trained on AutoDL NVIDIA RTX 5090 GPUs (32 GB VRAM) using PyTorch 2.6 and CUDA 12.8 with BF16 mixed precision. Both full decoders (E_0) and efficient baseline decoders (E_1) were trained for 45,000 steps with identical data augmentations, initial learning rate (0.001), and crop dimensions ($96 \times 96 \times 96$). Validation was conducted every 250 steps.

Baseline Comparison: Table 1 reports aggregate computational complexity and segmentation accuracy across all three architectures. The efficient decoder configuration (E_1) follows the parameter reduction strategy of EffiDec3D [7]. Across all backbones, E_1 achieves substantial computational savings (up to 91.4% FLOP reduction in 3D UX-Net) while preserving overall segmentation accuracy. For instance, MedNeXt-M-K3 E_1 achieves an 81.47% Mean Dice score (compared to 83.74% for E_0) while reducing inference latency from 40.9 ms to 18.7 ms.

Factorial Factor Analysis: To separate the effects of decoder channel dimension (C_{dec}) and spatial resolution scaling (RF), we conduct a 2×2 factorial ablation on 3D UX-Net (Table 2). Reducing decoder channels from 384 to 48 (C_{48}, RF_1) reduces parameters from 46.72M to 2.24M while maintaining a 79.80% Mean Dice. Omitting the high-resolution stage (C_{384}, RF_2) reduces inference latency to 16.1 ms. Combining channel reduction and stage removal (C_{48}, RF_2) yields the E_1 configuration (1.84M params, 49.49 GMac), confirming that channel scaling and resolution reduction act as distinct efficiency factors.

4.2. Spatial Heterogeneity of Decoder Contribution

We evaluate whether additional decoder capacity yields uniform prediction changes across volumetric space. For parameter-matched decoder pairs,

Table 2: 2×2 Factorial analysis of decoder channel reduction (C_{dec}) and resolution scaling (RF) on 3D UX-Net (BTCV dataset).

Configuration	C_{dec}	RF	Params (M)	FLOPs (GMac)	Latency (ms)	Mean Dice (%)
Full (C_{384}, RF_1)	384	1	46.72	657.76	43.3	80.44
Chan-Only (C_{48}, RF_1)	48	1	2.24	388.15	35.2	79.80
Res-Only (C_{384}, RF_2)	384	2	46.32	319.10	16.1	77.97
Effi (C_{48}, RF_2)	48	2	1.84	49.49	8.0	78.35

the global net benefit of full decoders remains statistically net-neutral across architectures. On the BTCV benchmark, the subject-mean net flip rate is $+0.046\%$ (95% CI: $[-0.0002\%, +0.0011\%]$) for 3D UX-Net, -0.0008% (95% CI: $[-0.044\%, +0.039\%]$) for SwinUNETR, and $+0.085\%$ (95% CI: $[+0.024\%, +0.152\%]$) for MedNeXt-M-K3. As shown in Figure 2, positive voxel corrections are counterbalanced by negative degradations across the global volume.

Spatially, boundary-level analysis demonstrates that this net-neutrality is driven by localized prediction shifts. The error rate near object boundaries is $3.93\times$ higher than in interior regions for 3D UX-Net and $4.67\times$ higher for MedNeXt-M-K3. Net flip activity across all three networks is concentrated within a narrow boundary band (0–1 voxels from organ edges).

Anatomical evaluation confirms this spatial heterogeneity across organ types (Figure 3). Small and complex structures show higher prediction volatility: the left and right adrenal glands exhibit positive flip rates of 16.2%/16.9% in 3D UX-Net, 14.4%/14.8% in SwinUNETR, and 18.3%/10.4% in MedNeXt-M-K3. Conversely, large organs such as the liver maintain low flip rates ($< 3\%$) across all models.

These results indicate that decoder interventions are localized to anatomical boundaries and smaller structures. Having established spatial heterogeneity, we next measure whether beneficial prediction flips are broadly distributed or spatially concentrated.

4.3. Spatial Concentration of Decoder Benefit

We evaluate the spatial concentration of positive net benefit across volumetric sub-blocks. Figure 4 shows the oracle coverage curves for 3D UX-Net, SwinUNETR, and MedNeXt-M-K3. Across all three architectures, an oracle selecting the top 5% of blocks recovers 80% of the full decoder's total positive net gain ($K_{80} = 5\%$). At a 10% volume budget, oracle coverage reaches 99.98%, reaching 100% at a 20% budget.

While the full decoder modifies predictions across broad spatial regions, net accuracy gains are restricted to 5% of the volume. In the remaining 95% of spatial blocks, additional decoder parameters generate redundant transformations without improving net correctness. This spatial concentration confirms that uniform decoder scaling introduces structured spatial redundancy.

4.4. Predictability of Decoder Utility

We test whether inference-time uncertainty signals (predictive entropy and confidence) from the efficient baseline can identify beneficial spatial regions prior to full decoder execution.

Across all backbones (Figure 5), we observe a divergence between predicting prediction instability and predicting net improvement:

- **3D UX-Net**: AUROC for flip location (instability) is 0.903 (95% CI: $[0.876, 0.926]$), whereas AUROC for flip direction (improvement vs. degradation) is 0.592 (95% CI: $[0.560, 0.626]$).
- **SwinUNETR**: Flip location AUROC is 0.912 (95% CI: $[0.884, 0.934]$), while flip direction AUROC is 0.561 (95% CI: $[0.512, 0.608]$).
- **MedNeXt-M-K3**: Flip location AUROC is 0.931 (95% CI: $[0.917, 0.944]$), while flip direction AUROC is 0.545 (95% CI: $[0.526, 0.567]$).

Because uncertainty fails to discriminate flip direction, the deployable selection gap remains zero. At a 20% execution budget, entropy-guided block selection yields net accuracy gains indistinguishable from a random selection baseline across all three networks. This demonstrates that identifying uncertain regions is insufficient for selective decoder execution, as routing decisions require directional utility estimation.

5. Discussion

This characterization study provides insights into how decoder capacity functions across spatial regions in 3D medical image segmentation. Rather than relying solely on global performance metrics, our analysis examines the spatial mechanisms governing decoder utility.

5.1. Mechanisms of Boundary Concentration and Net-Neutrality

Deep encoder backbones extract high-level semantic representations capable of segmenting homogeneous organ interiors. Consequently, additional decoder capacity provides minimal marginal benefit inside solid anatomical structures. At organ boundaries, however, voxels present class ambiguity and complex structural transitions. While full decoders attempt to refine these boundary voxels, static capacity produces a two-sided effect: positive corrections are offset by negative degradations on previously correct predictions, resulting in a net-neutral global balance ($R_{net} \approx 0$).

Standard evaluation metrics such as the Dice similarity coefficient average positive corrections and negative degradations across the entire volume, concealing this localized net-neutral behavior. This explains why global evaluation paradigms have historically favored uniform decoder scaling despite underlying spatial redundancy.

5.2. Uncertainty and Directional Utility

Predictive entropy measures epistemic difficulty and spatial ambiguity, effectively identifying where predictions are unstable near boundaries (AUROC > 0.90). However, entropy indicates region difficulty rather than whether a static decoder has the appropriate inductive bias to resolve the error. Because the efficient baseline is already competitive, executing a heavy decoder on uncertain voxels is equally likely to degrade a correct prediction as it is to fix an error (AUROC ≈ 0.55).

5.3. Architectural Implications and Scope Bounds

These spatial patterns are consistent across large-kernel CNNs (3D UX-Net), vision transformers (SwinUNETR), and ConvNeXt-style architectures (MedNeXt-M-K3), demonstrating that structured decoder redundancy is a general characteristic of U-shaped 3D segmentation models.

Consequently, theoretical efficiency gains cannot be realized simply by gating static decoders with uncertainty thresholds. Moving forward, dynamic decoder designs should allocate capacity based on feature complexity and directional utility rather than scalar uncertainty scores.

Finally, our scope is explicitly bounded: we characterize decoder computation under controlled interventions while holding encoders fixed. This focus isolates decoder dynamics without claiming that encoder representations are redundant or that U-Net architectures as a whole are unnecessary.

6. Conclusion

We presented a systematic characterization of decoder computation in 3D medical image segmentation architectures. By evaluating voxel-level prediction shifts between matched full and efficient decoders, we demonstrated that decoder utility is spatially heterogeneous, highly concentrated at organ boundaries, and net-neutral at the global level. Furthermore, we showed that while inference-time uncertainty reliably predicts where predictions change, it cannot predict whether additional capacity will improve or degrade accuracy.

These findings highlight the limitations of static decoding and naive uncertainty-based gating. Future work should focus on developing feature-adaptive decoders that dynamically route computation based on spatial utility rather than uniform resolution scaling.

CRediT authorship contribution statement

Author One: Conceptualization, Methodology, Writing – original draft. **Author Two**: Data curation, Software. **Author Three**: Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study (BTCV/Synapse, FeTA 2021) are publicly available.

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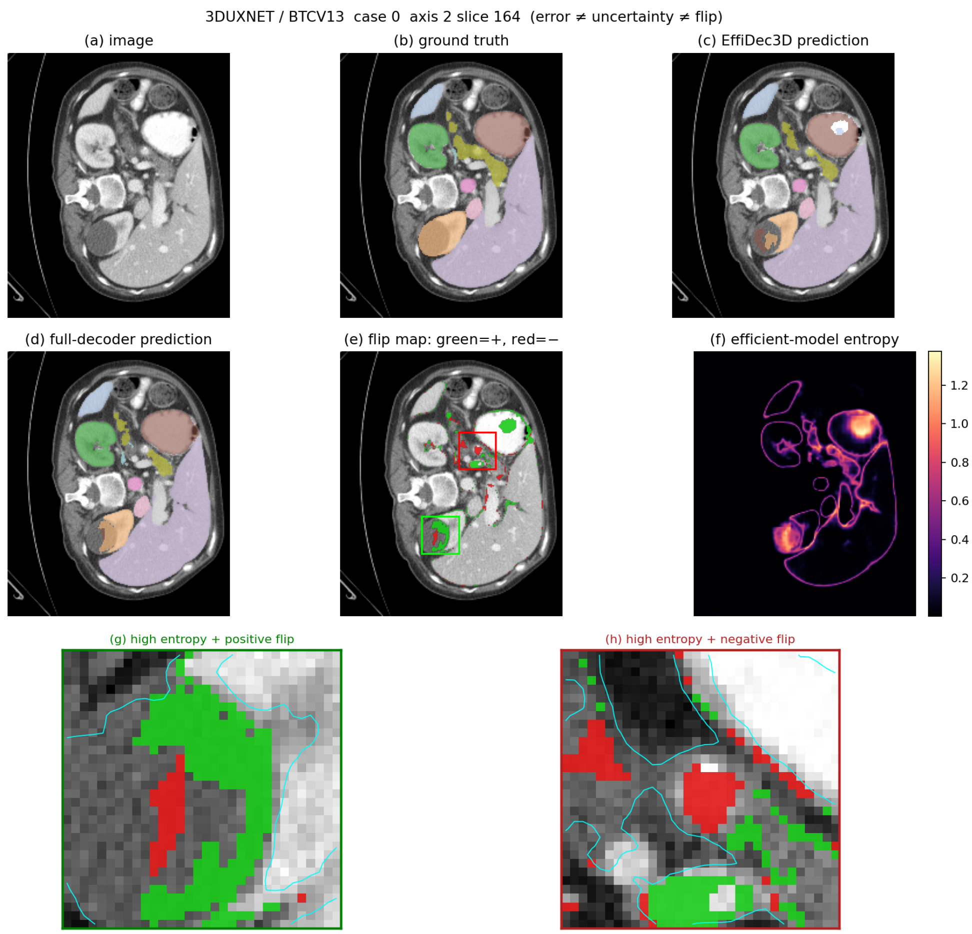
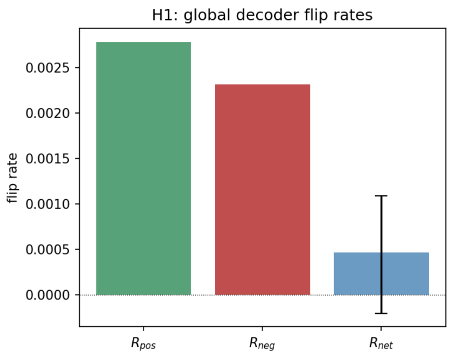
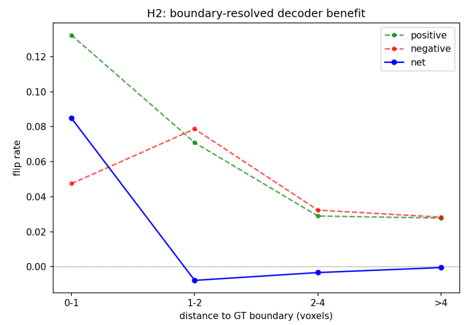


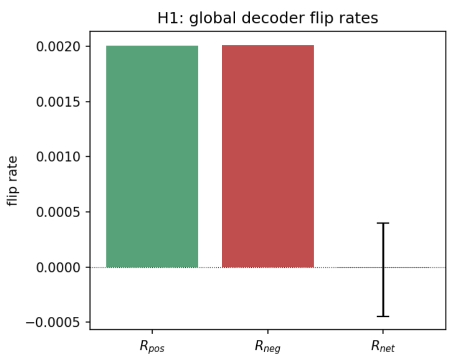
Figure 1: Motivation: Decoder computation produces spatially localized prediction flips. While the full decoder corrects certain errors of the efficient baseline (positive flips), it also alters previously correct predictions (negative flips) near high-entropy organ boundaries, demonstrating non-uniform spatial utility.



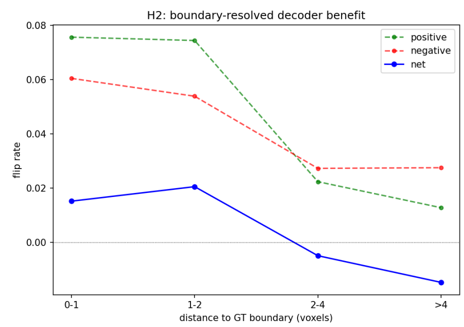
(a) 3D UX-Net Global Flips



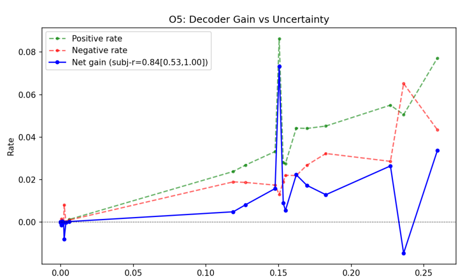
(b) 3D UX-Net Boundary Flips



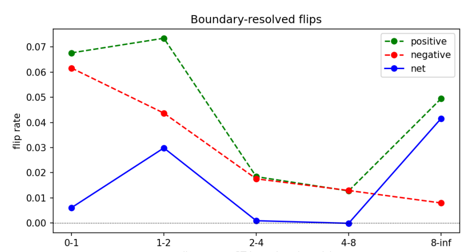
(c) SwinUNETR Global Flips



(d) SwinUNETR Boundary Flips

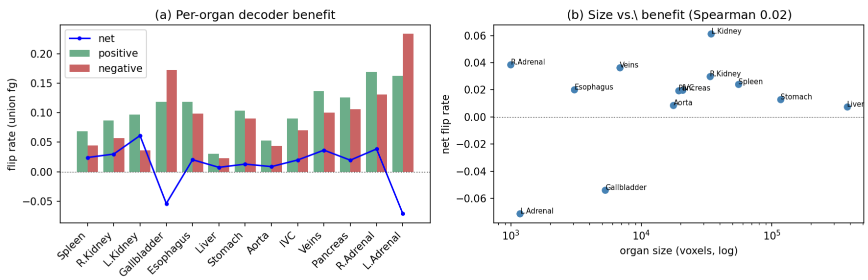


(e) MedNeXt-M-K3 Global Flips

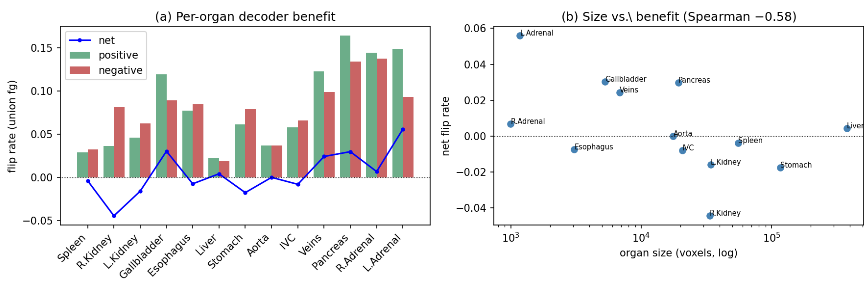


(f) MedNeXt-M-K3 Boundary Flips

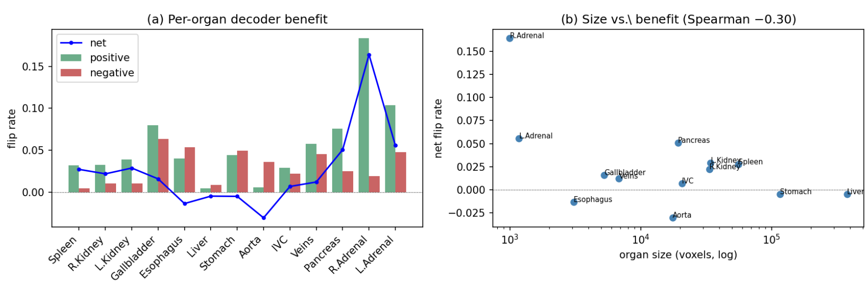
Figure 2: Voxel-level net benefit and spatial boundary concentration across architectures. Left Column: Global positive, negative, and net flip rates for (a) 3D UX-Net, (c) SwinUNETR, and (e) MedNeXt-M-K3. Right Column: Net flip rate as a function of distance to organ boundaries for (b) 3D UX-Net, (d) SwinUNETR, and (f) MedNeXt-M-K3.



(a) 3D UX-Net Organ-Level Flips

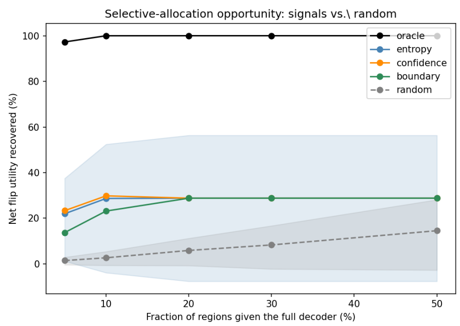


(b) SwinUNETR Organ-Level Flips

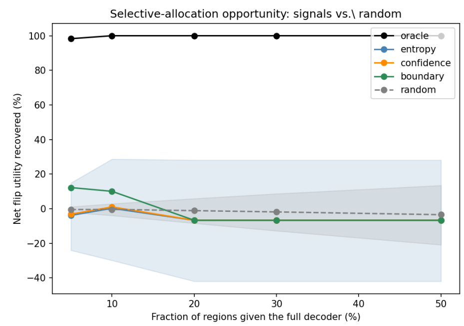


(c) MedNeXt-M-K3 Organ-Level Flips

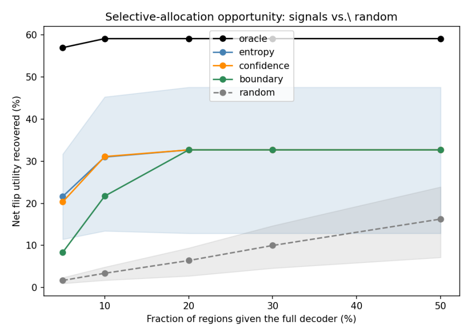
Figure 3: Organ-level flip breakdown across (a) 3D UX-Net, (b) SwinUNETR, and (c) MedNeXt-M-K3. Complex and small anatomical structures show high prediction volatility, whereas large organs remain stable.



(a) 3D UX-Net Concentration Curve

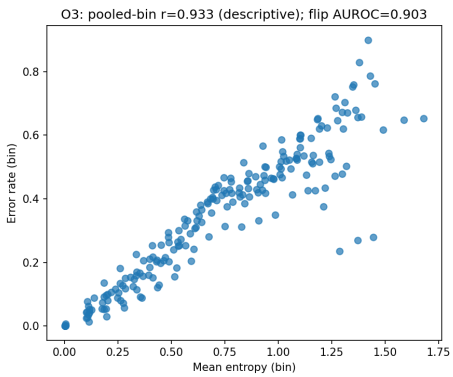


(b) SwinUNETR Concentration Curve

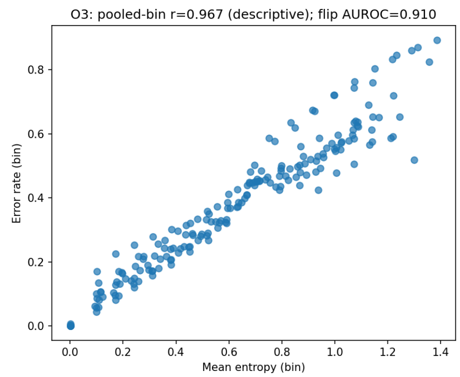


(c) MedNeXt-M-K3 Concentration Curve

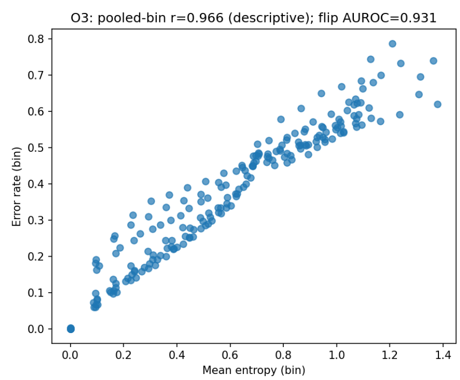
Figure 4: Oracle concentration curves for (a) 3D UX-Net, (b) SwinUNETR, and (c) MedNeXt-M-K3, showing that 80% of positive net benefit (K_{80}) is concentrated in 5% of spatial blocks.



(a) 3D UX-Net Predictability



(b) SwinUNETR Predictability



(c) MedNeXt-M-K3 Predictability

Figure 5: Inference-time flip predictability across (a) 3D UX-Net, (b) SwinUNETR, and (c) MedNeXt-M-K3. Predictive entropy reliably identifies flip locations (AUROC > 0.90) but cannot distinguish positive corrections from negative degradations (AUROC ≈ 0.55).